

Julien Chevallier

Assistant Professor

23 route de Langrune
14440 Douvres-la-Délivrande
✉ julchevall@gmail.com
🌐 <https://jucheval.github.io>
Né le 4 octobre 1989 à Nice



Updated on January 9, 2026.

Academic experience

- 2017 – ... **Assistant Professor**, *UFR IM²AG, Université Grenoble Alpes*, Grenoble
Lay-off since September 2025.
- 2016 – 2017 **Postdoc**, *AGM, Université de Cergy-Pontoise (UCP)*, Cergy-Pontoise
Supervisor : Eva Löcherbach.
Financed by Labex MME-DII.
- 2013 – 2016 **PhD Thesis**, *LJAD, Université Nice Sophia Antipolis (UNSA)*, Nice
Title : Modelling large neural networks via Hawkes processes.
Supervised by Patricia Reynaud-Bouret and François Delarue.
Defense: September 9th, 2016.
- 2012 – 13 **Postgraduated degree in Mathematics**, *ENS DE LYON*, Speciality: Probability
- 2011 – 12 **“Agrégation” in Mathematics**, *Competitive exam to qualify as high school teacher in Mathematics*, 36th over 308

Scientific research activity

The exhaustive list of my publications can be found in my [CV HAL](#).

Thesis (PhD and Habilitation)

- J. Chevallier : **Modelling large neural networks via Hawkes processes**, *Université Côte d’Azur* (2016).
Available on [tel-01415622v2](#).
- J. Chevallier : **A journey in the fields of PDE, probabilities and statistics with point processes**, *Université Grenoble Alpes* (2025).
Available on [tel-05431747](#). Code available on [this GitHub repo](#).

Published papers

- [P1] J. Chevallier et T. Lalöe : **Detection of dependence patterns with delay**, *Biometrical Journal*, 57:1110-1130 (2015). DOI : 10.1002/bimj.201400235
Available on [arXiv:1406.0476](#).
- [P2] J. Chevallier, M.J. Caceres, M. Doumic et P. Reynaud-Bouret : **Microscopic approach of a time elapsed neural model**, *Mathematical Models and Methods in Applied Sciences*, 25, 2669 (2015). DOI : 10.1142/S021820251550058X
Available on [arXiv:1506.02361](#).
- [P3] J. Chevallier : **Mean-field limit of generalized Hawkes processes**, *Stochastic Processes and their Applications* (2017). DOI : 10.1016/j.spa.2017.02.012
Available on [arXiv:1510.05620](#).
- [P4] J. Chevallier : **Fluctuations for mean-field interacting age-dependent Hawkes processes**, *Electronic Journal of Probability* (2017). DOI : 10.1214/17-EJP63
Available on [arXiv:1611.02008](#).
- [P5] J. Chevallier, A. Duarte, E. Löcherbach et G. Ost : **Mean Field Limits for Nonlinear Spatially Extended Hawkes Processes with Exponential Memory Kernels**, *Stochastic Processes and their Applications* (2018). DOI : 10.1016/j.spa.2018.02.007
Available on [arXiv:1703.05031](#).

- [P6] J. Chevallier : **Stimulus sensitivity of a spiking neural network model**, *Journal of Statistical Physics* (2018). DOI : 10.1007/s10955-017-1948-y
Available on [arXiv:1709.06958](#).
- [P7] J. Chevallier : **Uniform decomposition of probability measures: quantization, clustering and rate of convergence**, *Journal of Applied Probability* (2018). DOI : 10.1017/jpr.2018.69
Available on [arXiv:1709.06958](#).
- [P8] J. Chevallier et al. : **Network of interacting neurons with random synaptic weights**, *ESAIM: Proceedings and Surveys, EDP Sciences*, 55: 1037-1045 (2019). DOI : 10.1051/proc/201965445
Available on [hal-01928990](#).
- [P9] J. Chevallier et G. Ost : **Fluctuations for Spatially Extended Hawkes Processes**, *Stochastic Processes and their Applications*, 130 (9), 5510-5542 (2020). DOI : 10.1016/j.spa.2020.03.015
Available on [arXiv:1909.06401](#).
- [P10] J. Chevallier, A. Melnykova et I. Tubikanec : **Diffusion approximation of multi-class Hawkes processes: Theoretical and numerical analysis**, *Advances in Applied Probability*, 53(3), 716-756 (2021). DOI : 10.1017/apr.2020.73
Available on [arXiv:2003.10710](#).
- [P11] J. Chevallier, E. Löcherbach et G. Ost : **Inferring the dependence graph density of binary graphical models in high dimension**, Accepted in *Annals of Statistics*, (2025).
Available on [hal-04609972](#).

Submitted papers

- [S1] P. J. Gerrish, B. Galeota-Sprung, F. Cordero, P. Sniegowski, A. Colato, N. Hengartner, V. Vejalla, J. Chevallier, B. Ycart : **Natural selection and the advantage of recombination** (2020).
Available on [bioRxiv 2020.08.28.271486](#).
- [S2] P. J. Gerrish, B. Galeota-Sprung, P. Sniegowski, A. Colato, J. Chevallier, B. Ycart : **Natural selection promotes the evolution of recombination 1: between the products of natural selection** (2021).
Available on [bioRxiv 2021.06.07.447320](#).
- [S3] J. Chevallier : **Uniform in time modulus of continuity of Brownian motion** (2023).
Available on [hal-04361156](#).
- [S4] J. Chevallier et G. Ost : **Community detection for binary graphical models in high dimension** (2024).
Available on [hal-04803948](#).
- [S5] J. Chevallier, J.-F. Coeurjolly et R. Waagepetersen : **Critical point processes obtained from a Gaussian random field with a view toward statistics** (2025).
Available on [hal-05146209](#).

Open source software

- [L1] J. Chevallier : **MeanFieldGraph.jl** (2024). Available on [ce repo GitHub](#).
A Julia package devoted to the estimation of the connectivity parameter or the community detection in a latent (non sparse) high dimensional graph. Related to articles [P11] and [S4].
- [L2] J. Kling, R. Senne et J. Chevallier : **PointProcesses.jl** (2021). Available on [ce repo GitHub](#).
A Julia package devoted to simulation and estimation for temporal point processes.

Participation in conferences, seminars, workshops, etc.

International conferences – Invited speaker

- 2014/09/17 **PDE-MANS**, Grenade, Espagne
- 2015/09/09 **MathStatNeuro Workshop**, Nice, France
- 2017/06/20 **PDE-MANS 2017**, Grenade, Espagne
- 2018/04/13 **JSTAR**, Rennes, France
- 2019/06/18 **CIMPA Research Summer School**, La Havane, Cuba

2022/06/01 **New trends on Hawkes processes**, Toulouse, France

2022/06/24 **Rencontre autour du projet ANR ChaMaNe**, *Mini-course on neural field equations*, Île Rousse, France

International conferences – Contributed speaker

2015/06/02 **47es Journées de Statistique de la SFdS**, Lille, France

2015/06/09 **Congrès de la SMAI 2015**, *Mini-symposium "Processus de Hawkes et leurs applications"*, Les Karellis, France

2015/07/06 **30e European Meeting of Statisticians**, Amsterdam, Pays-Bas

2016/06/01 **ICMNS**, Antibes, France
Oral and poster presentations.

2017/12/13 **Deterministic and Stochastic Models in Neuroscience**, Toulouse, France

2018/06/12 **ICMNS**, Antibes, France

2018/06/14 **SPA 2018**, Göteborg, Suède

2024/05/28 **55es Journées de Statistique de la SFdS**, Bordeaux, France

2025/06/03-05 **56es Journées de Statistique de la SFdS**, Marseille, France

Supervision

Postgraduate level

- Research project – *CEMRACS* – 2017 (1 month, co-supervised with F. Delarue and E. Tanré)
Subject : Network of interacting neurons with random synaptic weights.
- **Kevin Polisano** – *Postdoc* – 2018/19 (1 an, co-encadré à 20% avec E. Gaussier et A. Leclercq-Samson)
Subject : Modélisation conjointe de dynamique de re-tweet et de follow pour twitter.
Become : CR CNRS en section 41 au LJK.
- **Anne-Lise Porté** – *PhD* – 2021/2022 (co-encadrée à 50% avec J.-F. Coeurjolly)
Subject : Processus ponctuel généré par les points critiques d'un champ Gaussien.

MSc level

- **Valentina Mazzi** – *M2 internship* – 2016 (5 months, co-supervised with P. Reynaud-Bouret)
Subject : Modélisation microscopique et macroscopique de neurones.
- **Marjolaine Demouge** – *M1 internship* – 2022 (co-supervised with J.-F. Coeurjolly)
Subject : Continuum de test statistique pour l'estimation de l'intensité d'un processus ponctuel spatial.
- **Martin Combelles** – *M1 internship* – 2025 (co-supervised with V. Brault)
Subject : Inférence du graphe d'interaction sur Twitter : Processus de Hawkes et modèle de blocs stochastiques.

BSc level

- Excellence internship – (6 weeks after final exams)
1 student (L1 MIN-Int) in 2023. 2 students (L3) in 2025.

Projects and contracts

- *IRGA IDEX* – STePP-Meteo – 2022/24 – 13,000€ – PI : Jean-François Coeurjolly
- *IRS IDEX* – GraTweet – 2017/19 – 15,000€ – PI : Vincent Brault
- *PEPS JCJC CNRS* – MaNHawkes – 2017/18 – 3,300€ – PI : Julien Chevallier

Scientific animation

2017 – 2022 **Co-organisation of DATA seminar**

2022 – 2025 **SVH team leader**, 13 permanent members.

Conference organisation

2019 **Co-organisation of congrès MathEnJeans 2019 de Grenoble**

- 2021 Co-organisation of BioHasard 2021, Online from 20/05 to 16/06 due to Covid-19.
- 2023 Co-organisation of SSIAB15, 24-26 May 2023.
- 2025 Co-organisation of GeoSto25, 23-27 June 2025.

Teaching activity

Teachings

- MSc level
 - Discrete time stochastic processes.
 - Temporal point processes.
- BSc level
 - Basics of descriptive and inferential statistics. Tutorial and practical classes.
 - Elementary analysis.

Popularisation

- Popular articles – *Images des Mathématiques* – 2016/17
Titre : Nos neurones se synchronisent-ils ? – [Part 1](#) – [Part 2](#)

Alternative teaching methods

- Cooperative teaching (JIGSAW)
This activity uses cooperative teaching to prove Perron-Frobenius theorem.

Collective responsibilities

Expertise assignments

- Member of a selection committee – MCF 26 à l'Université de Strasbourg – 2020
- Member of a selection committee – MCF 26 à l'ENSIMAG – 2022
- Reviewer for scientific journals : *Annals of Applied Probability* (4), *Acta Applicandae Mathematicae* (1), *Annales de l'IHP (B) Probabilités et Statistique* (3), *ALEA* (1), *Applied Probability Journals* (1), *ESAIM: Probability and Statistics* (1), *Electronic Communications in Probability* (1), *Electronic Journal of Probability* (2), *Journal of Approximation Theory* (1), *Journal of Mathematical Neuroscience* (2), *Journal of Statistical Physics* (1), *Journal of Theoretical Probability* (1), *Kinetic and Related Models* (2), *Scandinavian Journal of Statistics* (1), *Stochastic Processes and their Applications* (6), *Statistics and Probability Letters* (2), *Transactions on Modeling and Computer Simulation* (1).

Responsibilities

- 2015 – 2016 **Membre élu de la Commission Recherche de l'Université Nice Sophia Antipolis**,
Collège des étudiants
Présence à une séance plénière (2/3 heures) par mois pendant un an.
- 2019 – 2023 **Membre élu du conseil de l'UFR IM2AG**, *Collège des rangs B*
Participation à la commission recherche de l'UFR.

Various skills

Languages

French Native speaker
English Fluent

Computer skills

Programming Julia, L^AT_EX, R, Python
language